

Finite-candidate evidence contract

Actor-visible state
 $s_t^{\text{obs}}, z_e, Q_t$
 $m_{t,i}, \rho_{t,i}$

valid rows

Target-RR1 oracle
 M_{GT}, M_e, P_q
 $r_t^e, J_e^{(H)}, \Delta_{\text{look}}$

labels + headroom

One-step + rollouts
 $\hat{r}_{t,i}^e = f_{\psi}^{\text{VIN}}(s_t^{\text{cl0}}, z_e, q_{t,i})$
 random, greedy, lookahead, softmax

masked replay

Masked candidate-query Q_H
 $Q_{H,\theta}(s_t^{\text{cl0}}, z_e, q_{t,i})$
 $\arg \max_{i: m_{t,i} = 1} Q_{H,\theta}$

π_Q

Oracle re-evaluation
 $J_e^{(H)}, G_t^{(H)}, \eta_Q$
 cost, invalidity, coverage

after M5 evidence

M6+ bridge
 IQL, online Q_H
 actor-critic, simulators

Learned planner and evaluation